MCQ Question Paper

Double and Triple Integrals

1. Evaluate
$$\int_0^1 \int_0^2 (x+1)\,dy\,dx$$
a) 1
b) 2
c) 3
d) 4

2. Evaluate
$$\int_0^1 \int_0^x 2\,dy\,dx$$
a) $\frac{1}{2}$
b) 1
c) 2
d) 4

3. Evaluate
$$\iiint_{0\le x,y,z\le 1} 1\,dV$$
a) 0
b) 1
c) 2
d) 3

4. The integral
$$\int_0^2 \int_x^2 f(x,y)\,dy\,dx$$
represents
a) $0 \le y \le x \le 2$
b) $0 \le x \le y \le 2$
c) $x + y \le 2$
d) $0 \le x, y \le 2$

5. Evaluate
$$\int_0^1 \int_0^x y\,dy\,dx$$
a) $\frac{1}{6}$
b) $\frac{1}{4}$
c) $\frac{1}{3}$
d) $\frac{1}{8}$

6. If the order of integration is changed in a double integral:
a) Value changes
b) Value remains same if limits are changed correctly
c) Limits remain same
d) Integrand becomes zero

7. Evaluate
$$\iiint_{0\le x,y,z\le 1} (x+y)\,dV$$
a) 1
b) 2
c) $\frac{1}{2}$
d) $\frac{3}{2}$

8. Region bounded by coordinate planes and
$$x + y + z = 2$$
Correct integral:
a)
$$\int_0^2 \int_0^{2-x} \int_0^{2-x-y} dz\,dy\,dx$$
b)
$$\int_0^2 \int_0^2 \int_0^2 dz\,dy\,dx$$
c)
$$\int_0^2 \int_0^x \int_0^y dz\,dy\,dx$$
d)
$$\int_0^2 \int_0^{2-z} \int_0^{2-y} dx\,dy\,dz$$

9. Evaluate
$$\int_0^1 \int_0^{1-x} 1\,dy\,dx$$
a) $\frac{1}{2}$
b) 1
c) 2
d) $\frac{1}{4}$

10. Tetrahedron bounded by coordinate planes and
$$2x + y + z = 2$$
Correct order $dz\,dy\,dx$:
a)
$$\int_0^1 \int_0^{2-2x} \int_0^{2-2x-y} dz\,dy\,dx$$
b)
$$\int_0^2 \int_0^2 \int_0^2 dz\,dy\,dx$$
c)
$$\int_0^1 \int_0^x \int_0^y dz\,dy\,dx$$
d)
$$\int_0^2 \int_0^{2-y} \int_0^{2-z} dx\,dy\,dz$$

Answer Key

1. C
2. B
3. B
4. B
5. A
6. B
7. A
8. A
9. A
10. A